

RESEARCH ARTICLE | FEBRUARY 03 2025

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Physics of Fluids 37, 025101 (2025)

<https://doi.org/10.1063/5.0251192>



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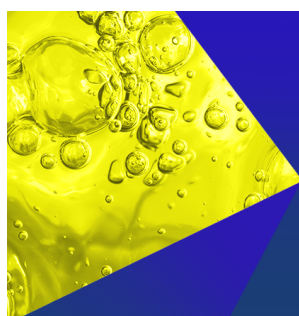
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Cite as: Phys. Fluids **37**, 025101 (2025); doi: [10.1063/5.0251192](https://doi.org/10.1063/5.0251192)

Submitted: 30 November 2024 · Accepted: 5 January 2025 ·

Published Online: 3 February 2025



View Online



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ABSTRACT

Recognizing the combustion mode in scramjet engines is critical for suppressing oscillations and stabilizing the combustion process in hypersonic aircrafts. Current accesses mainly depend on mechanical measurement and dominant frequencies based on image analysis methods, such as proper orthogonal decomposition and dynamic mode decomposition. However, these traditional methods either lack of precision or fall short of the need for prior knowledge, poor generalization, and low efficiency, posing challenges in practical implementations, especially when online controlling is highlighted in the scramjet combustions. Recently, machine learning (ML) has been introduced to the combustion community due to its superiority in high flexibility and efficiency in addressing complex problems. The classical convolutional neural network (CNN) architectures have been reported to achieve efficient combustion mode recognition in furnace combustion, swirling combustor, and rotating detonation engines. However, those CNN-based models are incapable of utilizing the global flame features and the coherences of local areas, resulting in insufficient accuracy and robustness in scramjet combustions with high inflow speed and distinct mode variations. To address this problem, this paper reports a Swin (shifted window) Transformer model, an advanced ML structure outstanding in capturing both global and local features by its self-attention mechanism with high computational efficiency, to identify combustion modes in scramjet engines. The Swin-T was trained and validated in a kerosene-fueled cavity-based scramjet combustor, and results show that it can achieve a considerable accuracy of 95.28%. Comparisons with CNN-based models further indicate that Swin-T outperforms in accuracy, efficiency, and robustness by around 0.7%, 80%, and 3%, respectively.

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I. INTRODUCTION

Technologies related to hypersonic aircrafts with the capability to operate at several times the speed of sound have been investigated with great effort during the last decades.¹ The supersonic combustion ramjet engine (scramjet) engine, as a promising candidate in the air-breathing propulsion system for hypersonic transport, is intricately related to the performance of the aircraft.^{2,3} In a scramjet engine, the propagation and stabilization of flame is a crucial and challenging problem as the oscillations and vibrations caused by combustion instability heighten the risk of aircraft structural damage and even bring catastrophic failure with huge economic loss.^{4–10} Different scramjet combustion modes with a large distinction in heat release and combustion strength are key constituents of the study in suppressing oscillations and stabilizing the combustion process.^{11–13} Therefore, it is necessary to develop an effective method that can recognize scramjet

combustion modes in order to reduce the risk caused by the oscillations of them.

Given its significance in the development of engines and its pivotal role in enhancing the safety of the system, in recent decades, significant attention has been directed toward the combustion modes of the scramjet combustor.¹⁴ In the research conducted by Micka and Driscoll, the definition of combustion modes for the scramjet combustor was established.¹⁵ Oscillations in the flame combustion modes within the reaction zone can be discerned. As a traditional way, several sensors are used in a scramjet combustor to monitor the condition of supersonic flame propagation. Wall pressure and stagnation exit temperature measurements were adopted by Aguilera and Yu to illustrate scramjet mode transition.¹⁶ Cai *et al.* also set some sensors, which could get wall-pressure distribution to conduct experiments on combustor mode transition.¹⁷ Although data from sensors can reflect the

tendency of flame mode transition, insufficient sampling rate and the use of averaged data result in measurement uncertainty, posing a threat to the accuracy of results. Image-based methods were also used in scramjet mode monitoring. In the research conducted by Nakaya *et al.*, CH* chemiluminescence sequential images captured by high-speed camera were used as the input of proper orthogonal decomposition (POD) method and dynamic mode decomposition (DMD) method to investigate combustion modes.² Although POD and DMD methods can detect scramjet flame structures by analyzing dominant frequencies, thereby achieving the goal of flame mode recognition, the accuracy of analyzing continuous large amounts of flame images is limited due to the difficulty in visually characterizing such structures and the lack of physical understanding of the structures reflected in the frequency domain signals. As a result, identifying features of scramjet combustion modes becomes challenging.

Machine learning (ML) approaches that are able to extract features directly from raw images, overcoming the limitations of conventional combustion diagnosis, have been adopted in the combustion field.¹⁸ Artificial neural networks (ANN) as widely used in flame analysis can extract combustion information and find relations between inputs and outputs by statistical methods automatically.¹⁹ Ren *et al.* developed a multi-layer perceptron (MLP) neural network method to reconstruct three-dimensional flame temperature with different equivalence ratios (ER), and the effectiveness of the method is verified through CO₂, H₂O, and CO mole fractions reconstruction.²⁰ Wang *et al.* adopted a two-step neural network to assist soot radiation-based thermometry for flame temperature field retrieval and the result indicated that the model had high accuracy in temperature prediction.²¹ Gao *et al.* used an artificial neural network to develop a pattern recognition model to differentiate the flame state of a scramjet. However, ANN needs prior knowledge, which depends on professional experience for feature extraction, and lacks generalization performance for unseen combustion.²² This approach also has the problem of poor robustness where shallow features are only suitable for specific diagnostic tasks. Inspired by artificial neural networks, a convolutional neural network (CNN) with superior performance in image processing has become a promising image-based method in combustion analysis.²³ Zhang *et al.* reported a conditional generative adversarial network-based convolutional operation to generate soot particles distribution in turbulent flames, achieving high accuracy in predicting planar laser-induced incandescence images.²⁴ Nie *et al.* presented a methodology of reconstructing flame temperature fields using CNN, and the results showed good statistical agreement with experiments.²⁵ Flow field reconstruction effectively reflects global information; however, these efforts typically focus on the reconstruction of specific metrics, such as temperature and velocity. Analyzing flame modes based solely on a single metric is insufficient. Choi *et al.*²⁶ developed a 2D CNN mode to enhance the accuracy of combustion state prediction. Ghosal *et al.*²⁷ formulated combustion instability detection as binary classification using 3D CNN architecture. However, works that utilize CNNs solely as feature extraction tools overlook the complexity and diversity of flame characteristics. Due to their limited capacity for long-term dependencies, many flame features may be neglected. Gangopadhyay *et al.*²⁸ combined CNN architecture and long short-term memory (LSTM) to build an enhanced framework to recognize binary combustion modes. Lyu *et al.*²⁹ investigated and improved the CNN-LSTM architecture to deal with a similar task. These networks

address the long-term dependency issue; however, due to the weight-sharing mechanism of CNNs, all flame features are assigned equal weight. In reality, the importance of certain features can be reduced for specific problems, thereby enhancing the ability to capture more significant features.

The scramjet combustion has severe inflow conditions, stagnation temperature, and stagnation pressure. In this extreme scenario, the description of flame combustion requires the observation and analysis of the overall structure of the flame, indicating that the extraction of large-scale image information is essential. However, CNNs have been proven that they have worse performance in large-scale image processing tasks comparing with the transformer-based methods³⁰ and industrial³¹ images, indicating high generalization performance. The embedding and self-attention mechanism in transformer shows strong capability to address the long-term dependency issue and can “pay more attention” to important features. Vision Transformer (ViT) architecture, as the first network which adopts a transformer neural network as the backbone to combat image processing issues, has posed a challenge to the conventional convolution-based backbone structures in the domain of image processing.^{30,32} This is attributed to factorized self-attention mechanism in the ViT model for finding spatial relationships among non-overlapping image patches, thereby effectively balancing speed and accuracy.³³ It has been validated that the adoption of such novel architecture substantially reduces model complexity while preserving nearly equivalent performance levels.³⁴ However, the Vision Transformer (ViT) cannot facilitate communication between the information of image patches, whereas convolutional neural networks (CNNs) can combine information from different regions by using convolutional kernels with specific strides, thereby extracting features from images. This issue has been addressed by the Swin Transformer (Swin-T) architecture, using an improved self-attention mechanism: shifted window attention mechanism on image patches to get local and global information.³⁵ By mitigating the performance drawbacks inherent in traditional approaches reliant on direct upsampling or deconvolution (like CNN does) for image processing, Swin-T achieves a heightened level of versatility and strikes a refined balance between recognition speed and accuracy in the realm of image recognition tasks.^{35–37}

In this paper, a Swin Transformer neural network is proposed to conduct mode recognition of scramjet combustion flames. Experiments were carried out in a kerosene-fueled scramjet combustor by a multi-channel gliding arc (MCGA) plasma to capture flame images under four different combustion modes. The structure of Swin-T is optimized by adjusting hyperparameters. The performance of the model utilized is compared with ViT and other CNN-based models. Additionally, by adding different levels of noise, the robustness of the proposed model is tested. Results obtained from the validation and evaluation are presented and discussed.

II. EXPERIMENT

A. The experimental settings

1. The scramjet combustor

Figure 1 shows the structure of the scramjet combustion chamber, including the multi-channel gliding arc (MCGA) igniter. Optical access is provided through a quartz glass window with a size of 60 × 200 mm² mounted on the sidewall. The combustor

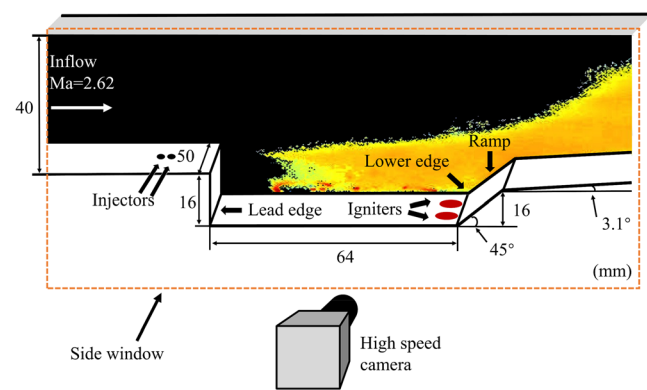


FIG. 1. Scramjet cavity with the MCGA plasma.

consists of three main parts: a cross-sectional isolator, a supersonic cavity combustor, and an expansion section. The entrance isolator of the combustor was designed with dimensions of 50 mm (width) $\times$ 40 mm (height), and the cavity measures 64 mm (length) $\times$ 50 mm (width) $\times$ 16 mm (height) with a 45° back-ramp. Two MCGA plasma igniters were positioned near the back ramp of the cavity, with their edges coinciding with the lower edge of the back ramp. The igniters, equipped with a six-channel gliding arc, were used to enhance combustion. The breakdown distance for each channel of the gliding arc is 3 mm between the electrodes and the iron wall. The expansion angle was set at 3.1° to enhance the circulation of the primary vortex, thereby reducing the residence time within the cavity. The Laval nozzle was designed for $Ma = 2.92$ (simulating flight at $Ma = 6$) with $T_t = 1650$ K and $P_t = 2.6$ MPa. Four vertical fuel injectors, each with a diameter of 1 mm, were placed upstream of the leading edge of the cavity. Details of the experiment can be found in Refs. 3 and 38–43.

2. The diagnostic techniques

During the process of obtaining sequential flame images, a high-speed camera (SA-Z), a high-speed camera intensifier (UVi model 1850), and an OH narrowband filter (310 $\pm$ 10 nm) to perform OH* chemiluminescence visualization of the supersonic combustion. The SA-X2 had a frame rate of 20 kHz and an exposure time of 48 μ s, triggered by an oscilloscope. The MCGA plasma is produced by a plasma generator (CTP-2000 K, designed by Suman technology) achieved by a multiple-discharge circuit. The discharge waveforms were measured by a high-voltage probe (P6015A) and a current probe (TCP0030A). In addition, the emission spectra were recorded by a spectrometer (Ocean Optics) for characterizing the radicals generated by the MCGA discharge in the air flow.

TABLE I. Experimental conditions for the flame datasets.

Flow rate	Equivalence ratio	Pressure (MPa)	Mach number	Total incoming air temperature (K)	Number of frames
$m_k = 29.2$	ER = 0.59	2.6	2.62	1650	2402
$m_k = 21.7$	ER = 0.43	2.6	2.62	1650	2422
$m_k = 12.2$	ER = 0.24	2.6	2.62	1650	2380

B. The experimental dataset

Three kerosene-fueled scramjet combustion experiments were conducted under the three conditions listed in Table I, namely, with fixed $Ma = 2.62$, $T_t = 1650$ K, and $P_t = 2.6$ MPa while both kerosene mass flow rate m_k and equivalence ratio (ER) are set as variables. Each row in Table I describes the flame conditions, specifically the different settings of m_k and ER, as well as the number of flame images obtained during that experiment. There were 7204 snapshots of flames acquired in total, which were then labeled into four combustion modes following the categorization method in Ref. 3 and shown in Fig. 2(a). The intermediate mode includes flame images captured during flame blow-off as well as images taken when no fuel was injected. These images represent the transition process of the flame from absence to presence. Hence, the high-intensity signals in these images are caused by the gliding arc plasma rather than combustion-generated signals. In contrast, jet-wake stabilized mode combustion has a thick reaction zone and the position of the leading edge (POS) of the reaction zone is anchored near the injectors, which are located upstream of the leading edge. POS of both cavity shear layer mode and cavity stabilized mode is in the positive direction of inflow. The difference between these two modes is the signal in the cavity. To be specific, cavity shear layer mode has almost no reaction in the cavity, whereas the reaction zone of cavity stabilized mode almost fills the cavity.

Figure 2(b) shows the typical cases of the labeled combustion modes, the recording area is -60 to 150 mm in length and -20 to 80 mm in height. In each mode, two typical flame cases are shown. The original image size was 896×368 in pixel², which is equivalent to a spatial resolution of 0.27 mm/pixel, while to facilitate the DL processing and alleviate the computation burden, the recorded images were converted into a size of 224×224 in pixel². It is worth highlighting that such a preprocess did not affect the relative radicals coordinates and flame shapes the category rules essentially relied on, and thus, it would not influence the later procedures of the combustion mode recognition. Then, the image data in each mode were divided into training and testing sets by a ratio of 8:2.

III. METHODOLOGY

A. Swin Transformer model

Swin-T is an improved Transformer model and it can establish long-range dependencies by using moving windows. Figure 3 illustrates the architecture of the Swin Transformer network. Following Ref. 35, it takes a three-channel image with a size of $H \times W$ as input, transferring the image into non-overlapping feature patches through a patch partition module. It also explores feature coherences by four Swin Transformer stages, thereby targeting a specific combustion mode according to the implied feature relations extracted by the window shifted self-attention mechanism. A linear embedding module achieved by a hidden fully connection layer was incorporated to map

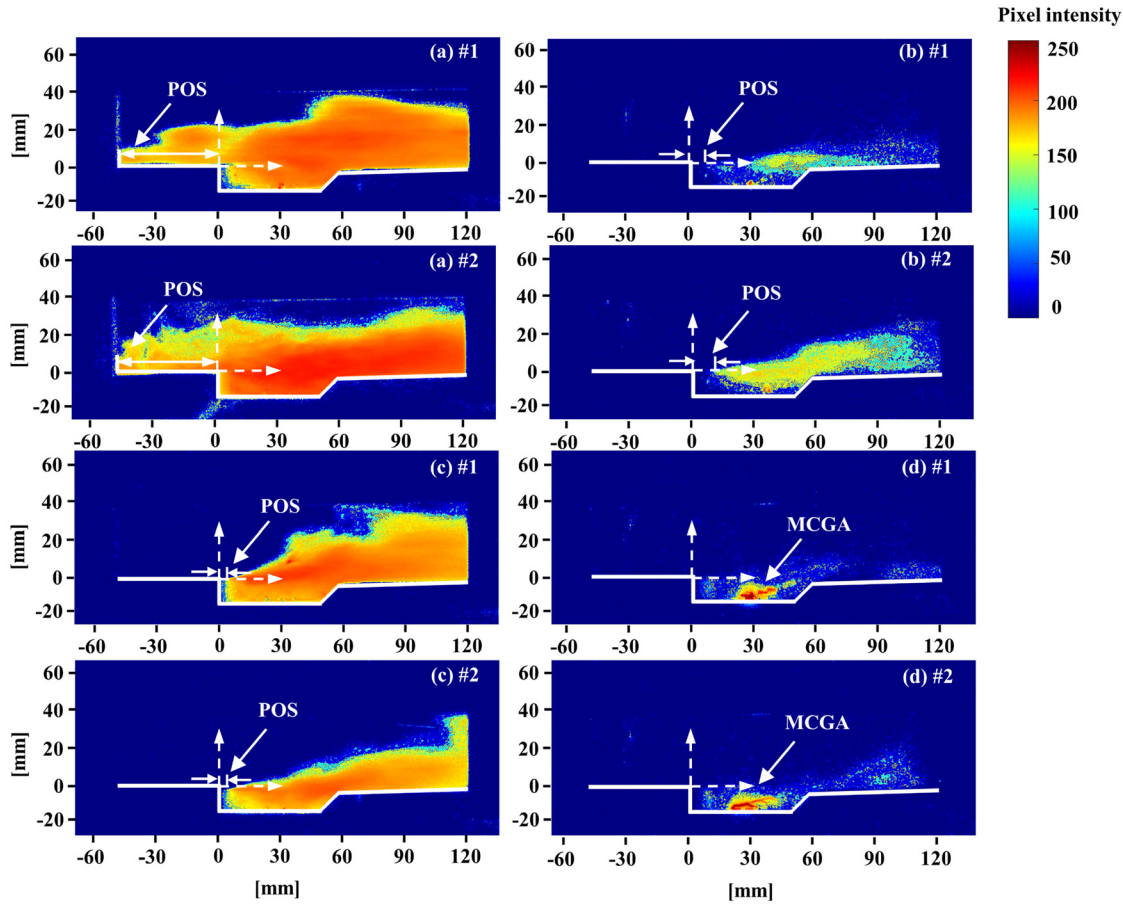


FIG. 2. Example of flame images under four different scramjet combustion modes, i.e., (a) jet wake stabilized mode, (b) cavity shear layer mode, (c) cavity stabilized mode, and (d) intermediate mode. In each mode, two typical flame images are presented.

the local patches to a dimension of $(H/4 \times W/4 \times C)$. C is a hyperparameter parameter set as 96 in this research. As shown in Fig. 3(a), the Patch Merging module fused adjacent features and halved the image width and height while doubling the channel, thus reducing the resolution of the feature image by downsampling.

The architecture of two successive transformer blocks is shown in Fig. 3(b). The Swin-T block is a cascade of two multi-head attention modules, consisting of window multi-head self-attention (W-MSA), shifted window multi-head self-attention (SW-MSA), and multi-layer perceptron (MLP). Its implementation can be formulated as follows:

$$\hat{z}^l = W - \text{MSA}(\text{LN}(z^{l-1})) + z^{l-1}, \quad (1)$$

$$z^l = \text{MLP}(\hat{z}^l) + \hat{z}^l, \quad (2)$$

$$\hat{z}^{l+1} = \text{SW} - \text{MSA}(\text{LN}(z^{l-1})) + z^l, \quad (3)$$

$$z^{l+1} = \text{MLP}(\hat{z}^{l+1}) + \hat{z}^{l+1}. \quad (4)$$

The window-based self-attention (W-MSA) and shifted window self-attention (SW-MSA) are based on the MSA. Their unique designs enable the model to learn local features within image patches. The image is divided into local windows, each containing a certain number

of patches. Attention values of local windows are computed using the MSA to extract features within the image. W-MSA computes attention within local windows. SW-MSA establishes connections between windows, enabling the representation of image feature information across windows. W-MSA and SW-MSA employ different window partitioning schemes, as illustrated in Fig. 3(b). As shown in the figure, W-MSA uses a traditional window partitioning scheme. SW-MSA, based on W-MSA, shifts the window rightward and downward by two patches, thereby generating a new window partitioning. The blank areas generated by the shifting windows are filled with patches of the corresponding color, which is shown in Fig. 3(b). By moving the window and utilizing the information provided by W-MSA, SW-MSA establishes connections between the original windows.

B. Loss function

W-MSA and SW-MSA are variants of image filters that extract specific information to characterize features in flame images, as reflected by the intensity relationships between pixels. The weights of W-MSA and SW-MSA play a crucial role in this process. Proper weight settings ensure that the processed images, regardless of flame

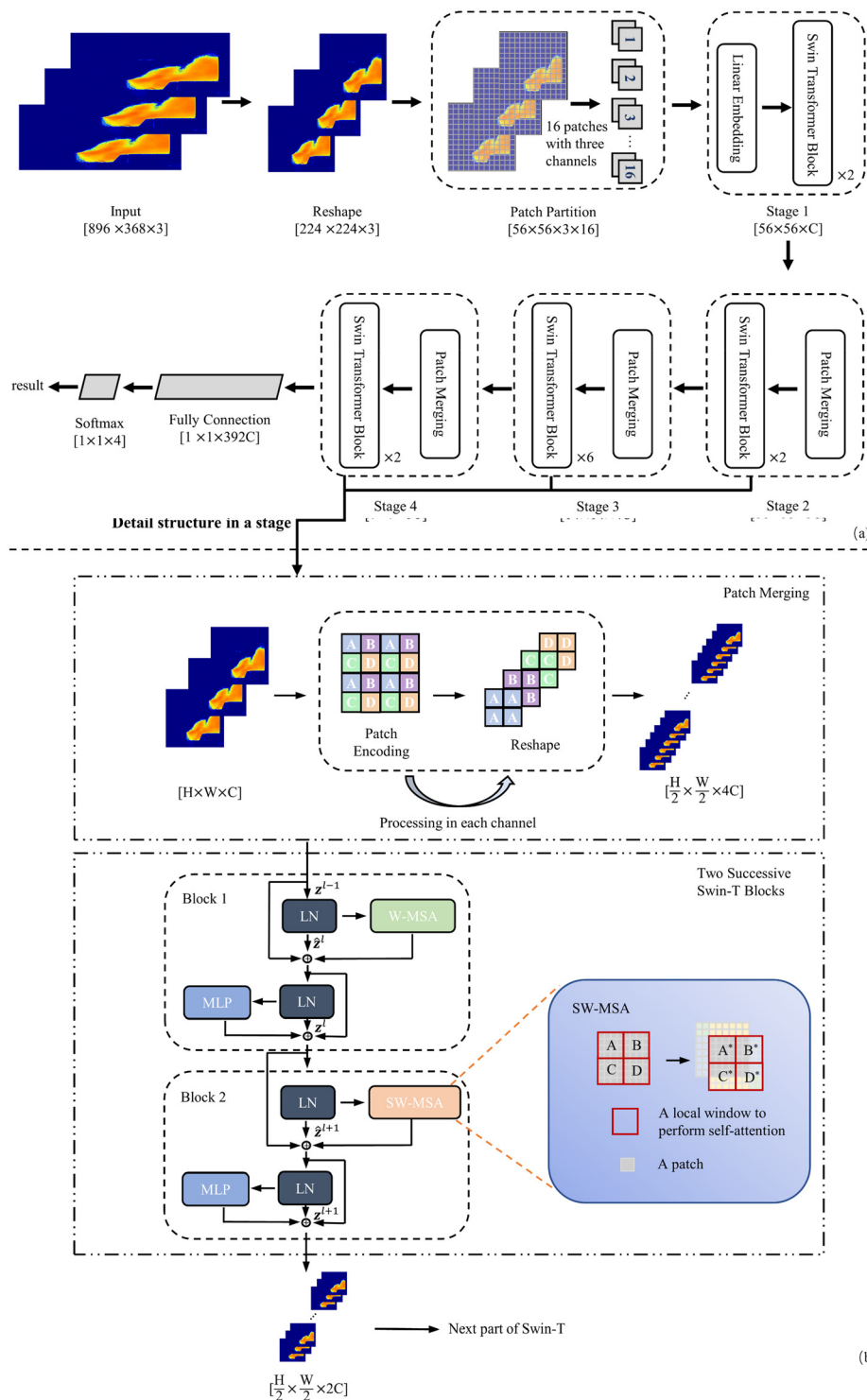


FIG. 3. The Swin Transformer architecture developed for scramjet combustion mode recognition: (a) the whole process of mode recognition and (b) detailed structure in a typical Swin-T stage.

modality, effectively capture the unique characteristics of each modality. Thus, during training, the weights must be iteratively updated until they meet the desired requirements, a process facilitated by the use of a loss function.

Following the previous classification models,^{29,44} the softmax function was adopted to convert the output feature matrix to categorizing probabilities before computing the loss functions. The feature matrix $\mathbf{Z}$ can be described as follows:

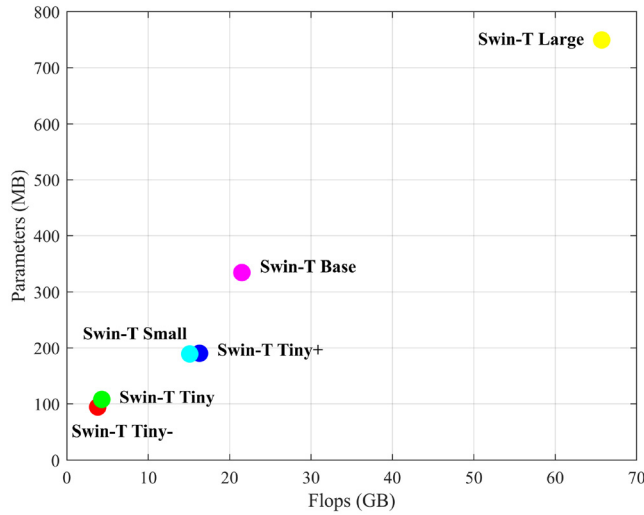


FIG. 4. Parameters and flops of different Swin-T architectures.

$$\mathbf{Z} = \mathbf{W}\mathbf{X} + \mathbf{b} = \begin{bmatrix} z_{1,1} & z_{2,1} & \cdots & z_{M,1} \\ z_{1,2} & & & z_{M,2} \\ \vdots & & \ddots & \\ z_{1,N} & z_{2,N} & \cdots & z_{M,N} \end{bmatrix}, \quad (5)$$

where $z_{i,j}$ represents the unnormalized logits for the i th sample being predicted as the j th class. $\mathbf{W}$ represents the weight matrix and $\mathbf{X} = \{x_1, x_2, \dots, x_M\}$ stands for the input vector with all flame images. $\mathbf{b}$ is a bias term that can also be updated during the training process. N represents the total number of categories. Since there are only four modalities in this study, the value of N is set to 4. M represents the number of images that need to be predicted. The probability of the i th mode p_{ij} can be obtained using the softmax function

$$p_{ij} = \text{softmax}(z_{i,j}) = \frac{\exp(z_{i,j})}{\sum_{k=1}^N \exp(z_{i,k})}, \quad (6)$$

where p_{ij} represents the predicted probability that the model assigns the i th sample to the j th class. Then, the cross-entropy loss was adopted as the loss function to train the model.

The cross-entropy loss function⁴⁵ is widely used in deep learning for image classification tasks. According to previous studies,^{29,44} it has been successfully applied to flame image classification tasks and the loss function can be defined as follows:

$$L = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^C y_{ij} \log(p_{ij}). \quad (7)$$

In this function, N represents the number of samples, also known as the batch size, while C denotes the number of categories. The variable y_{ij} indicates the actual label of the i th sample, where the j th class is marked as 1 if it is the correct category and 0 otherwise.

Next, the weights are updated through backpropagation to minimize the output of the loss function, which reflects the difference between the predicted labels and the actual labels. According to the chain rule, the gradient of the loss function with respect to the model weights can be expressed as follows:

$$\mathbf{g}_t = \frac{\partial L}{\partial \mathbf{W}} = \frac{\partial L}{\partial \mathbf{Z}} \frac{\partial \mathbf{Z}}{\partial \mathbf{W}}, \quad (8)$$

where L represents the loss function and $\mathbf{W}$ represents the weight matrix. During the training process, SGD optimizer⁴⁶ of stochastic optimization method is used to minimize cross-entropy

$$\mathbf{W}_{t+1} = \mathbf{W}_t - \eta \mathbf{g}_t, \quad (9)$$

where η represents the learning rate, which affects the calculation step. Through these steps, the weights are iteratively updated until the model's loss value converges, indicating that the difference between the predicted and actual values is minimal, meeting the requirements for flame mode recognition.

C. Swin-T model optimization

To evaluate an appropriate network architecture, a comparative experiment using a Swin-T-based model architecture was conducted. Swin-T has two hyperparameters: the number of Swin Transformer blocks in stage three, also known as the Layer number, and the Projected dimension (C) mentioned previously. The window size is set to $M=7$ by default. The query dimension of each head is $d=32$, and the expansion layer of each MLP is $\alpha=4$, for all experiments. The parameters and FLOPs of tested Swin-T architectures are shown in Fig. 4. These architectures were tested on the labeled data, and the results are presented in Table II. Increasing either the Layer number or C will result in a larger number of model parameters and more FLOPs, implying that larger networks have more stringent computational resource requirements. Compared to the increase in the number of layers, the increase in the projected dimension has a more significant impact on the model size and FLOPs. This is because the projected dimension affects all stages of the entire model, leading to an overall increase in model size. In contrast, the increase in the number of layers only impacts stage 3 in Fig. 3, resulting in the enlargement of only a part of the model.

TABLE II. Performance of the Swin-T model under different architecture.

Case	Swin-T tiny–	Swin-T tiny	Swin-T tiny+	Swin-T small	Swin-T base	Swin-T large
Layer number	4	6	6	18	18	18
Projected dimension (C)	96	96	128	96	128	192
Test accuracy	91.90%	93.32%	91.90%	93.63%	93.08%	94.01%
Inference time ($\times 10^{-3}$ s)	2.270	2.640	2.556	4.865	5.062	5.225

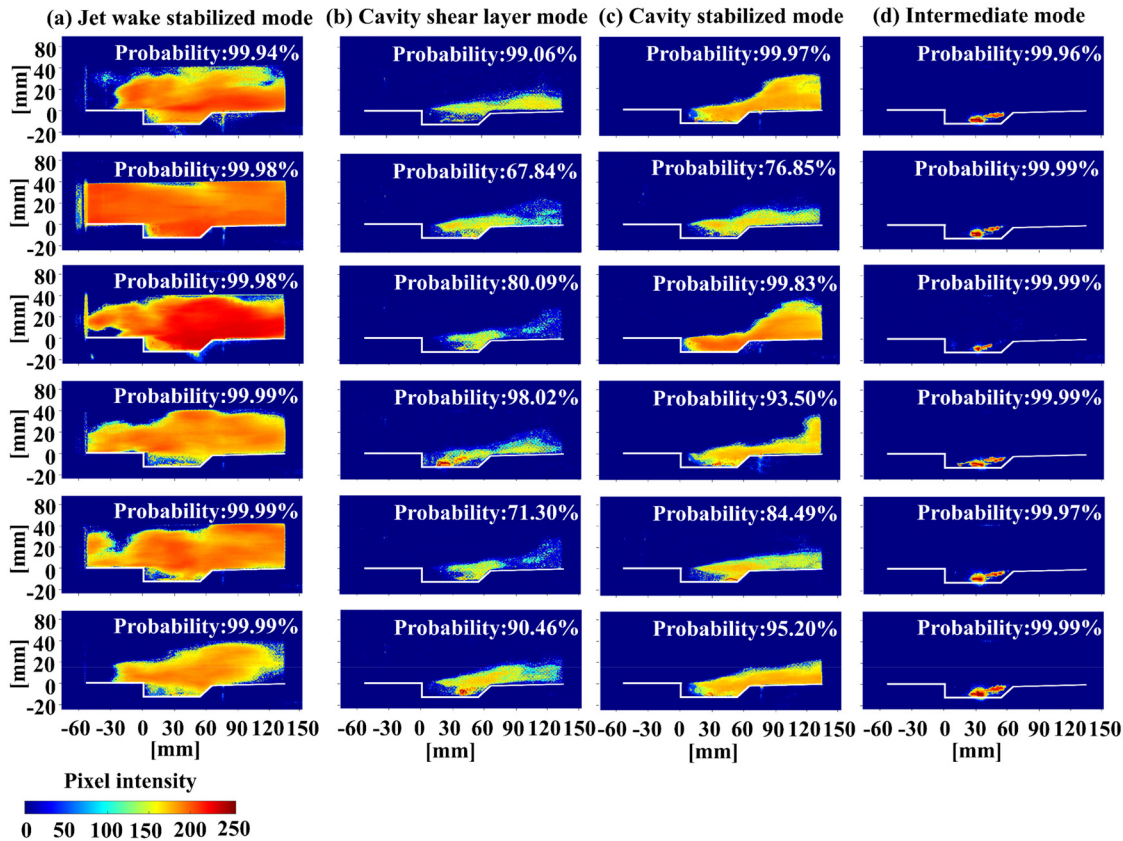


FIG. 5. Scramjet combustion flame image mode recognition with the probability from Swin-T predictions: (a) jet-wake stabilized mode, (b) cavity shear layer mode, (c) cavity stabilized mode, and (d) intermediate mode.

As can be seen from Table II, a smaller number of layers weakens the model's feature extraction capability, resulting in lower accuracy in testing. The reason is that the number of layers affects the number of times W-MSA and SW-MSA are processed. With multiple attention mechanism-based processes, the features of the image are more easily captured by the model. However, increasing the projected dimension without changing the number of layers can negatively impact the model's feature extraction, thereby reducing the accuracy. This is because the projected dimension affects the number of feature dimensions and the number of heads in the attention process. When the number of MSA processing is insufficient, it becomes challenging to extract features from high-dimensional data. When the number of layers is increased, thereby increasing the number of MSA processing iterations, the accuracy improves. This phenomenon confirms the validity of the previous analysis. Although the accuracy improves, the increase is modest and is accompanied by a significant increase in computational resource consumption.

To evaluate the real-time performance of the model, inference time that corresponds to the time from loading image into the model to making a decision (identifying the modality) is utilized and the results are shown in Table II. Notably, this metric is dependent on the performance of the hardware employed. In the tested environment, an RTX 2060 GPU was utilized. The experiment involved used

randomly generated $224 \times 224 \times 3$ vectors (shape of an flame image) as input to measure the inference time. Each individual model underwent 300 repeated experiments. To account for hardware warm-up, the first 50 iterations were discarded, and the average value of the subsequent 250 iterations is presented in the Table II. As observed from the table, larger models (as illustrated in Fig. 4) exhibit longer inference times, accompanied by relatively higher accuracy. This is attributed to the increased model depth, which involves more complex processing steps, such as multiple applications of the SW-MSA. The increased computational complexity leads to longer inference times. Notably, the Swin-T Tiny model achieves an excellent balance between inference time and accuracy. With a processing speed of approximately 2 ms/image, it is well-suited for practical turbulent flame mode recognition tasks.

IV. RESULTS AND DISCUSSION

This section gives quantitative analyses of the model performance, including accuracy, robustness, and ablation studies. In addition, the ViT, VGG, ResNet, and MobileNet models were also tested as comparisons. The proposed Swin Transformer was trained by the SGD optimizer,⁴⁶ and the learning rate was initially set as 0.001 with a cosine annealing schedule adopted. Model training was deployed on one Nvidia Tesla V100 GPU lasting 246 min.

A. Model accuracy

Figure 5 shows the snapshots of the recognized modes by the proposed Swin Transformer model along with the softmax calculated probabilities as confidence. It can be seen that the model exhibits probabilities of approximately 90.00% across all four modes in most shown cases, demonstrating its powerful capability of capturing flame features and mapping them to correct categories. Among the four modes, the intermediate mode can be successfully recognized with the highest probability of 99.99%, mainly because flames at this stage shrink to a minimal scale and present simple image features that can be easily captured by the model. The combustion in jet-wake stabilized mode was stably maintained with flames expanded throughout the combustor and featuring shape variations, and thus, it yielded slightly lower confidence level of around 99.94%. Since flames in both cavity shear layer mode and cavity stabilized mode developed in similar areas, they exhibit close features in integral shape and intensity in some cases as shown in Figs. 5(b) and 5(c). This causes recognition confusion and leads to a decline in recognition confidence in those two modes. Some very similar structures will cause a serious drop in confidence probabilities. For example, the probability of the two images in the middle row drops to around 70%.

B. Statistical analysis

Following previous recognition works,^{24,47} the standard accuracy, recall, and precision testing were used as the indices to evaluate the model performance. These indices can be described as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (10)$$

$$\text{precision} = \frac{TP}{TP + FP}, \quad (11)$$

$$\text{recall} = \frac{TP}{TP + FN}, \quad (12)$$

where TP is the number of true positives, FP is the number of false positives, FN is the number of false negatives, and TN is the number of true negatives.

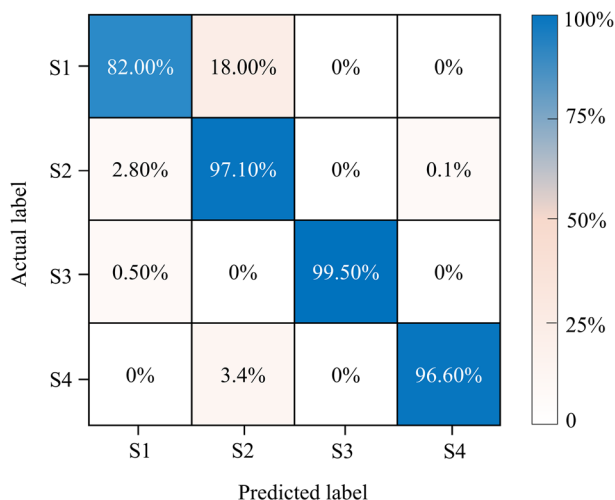


FIG. 6. The confusion matrix for the recognized modes. Data were calculated on 1440 images.

Figure 6 illustrates the recognition accuracy of the Swin-T model for different flame modes. In this figure, rows correspond to the predicted category, and columns represent the actual category. Consequently, the sum of the numbers in each row equals 100%, and the percentage along the diagonal indicates the proportion of correctly identified flame modes. It can be observed that the recognition accuracy for the Intermediate mode is 99.5%, which is the highest among the four categories. Given that this mode solely comprises plasma signals, it can be inferred that the model possesses a robust capability in distinguishing between plasma and flame. The recognition accuracies for the cavity stabilized mode and the jet-wake stabilized mode are relatively close, being 97.1% and 96.6%, respectively, both achieving high levels of performance. Notably, despite the cavity stabilized mode being rarely misidentified as the jet-wake stabilized mode by the model (with only 0.1% of such occurrences), the jet-wake stabilized mode is less frequently misclassified as the cavity stabilized mode. This suggests that the model is adept at capturing the combustion characteristics unique to the cavity stabilized mode. The confusion in modal recognition between these two modes, which share similarities in flame structure with flames primarily concentrated in and behind the cavity, underscores the intricacies involved in distinguishing between them. The recognition rate for the cavity shear layer mode is the lowest among the four flame modes, with errors primarily occurring in its misidentification as the cavity stabilized mode. This phenomenon can be attributed to the oscillations in the flame, which can lead to transitions between these two modes, as reported by Feng *et al.*³ During these transitions, the characteristics of the cavity shear layer mode gradually become blurred, with the flame width gradually increasing in the reaction zone. This makes it challenging for the model to capture the majority of the flame features that occur during the modal transition, resulting in confusion between the flame modes.

Figure 7 presents the precision and recall of Swin-T under the given dataset. The model's precision has reached 95.37%, with a recall of 93.74%, indicating that the model exhibits excellent performance in the recognition of supersonic combustion flame modes. The data presented in this figure lead to similar conclusions as those analyzed earlier. First, the intermediate mode exhibits the best performance in both evaluation metrics, with the highest recognition accuracy. The jet-wake stabilized mode follows closely, with its recall value slightly lower than that of the intermediate mode, contributing to its second-best performance. The worst-performing flame mode is the cavity shear layer mode, which exhibits the lowest scores in both precision and recall metrics. As previously mentioned, this can be attributed to the blurred features caused by the transitions between flame modes. In contrast, the cavity stabilized mode and the jet-wake stabilized mode demonstrate similar performance, with both exceeding 90% in both evaluation metrics.

C. Model comparison

In this work, four models with distinct network architectures were compared with Swin-T, and their test loss curves are illustrated in Fig. 8. As can be observed from the figure that, Swin-T achieves the lowest loss value while also maintaining a favorable convergence speed. Specifically, the loss value of Swin-T stabilizes around eight epochs, indicating that the model parameters have converged effectively. Among the models compared, only VGG16 exhibits a performance similar to Swin-T. Despite having a final loss value that is quite close to

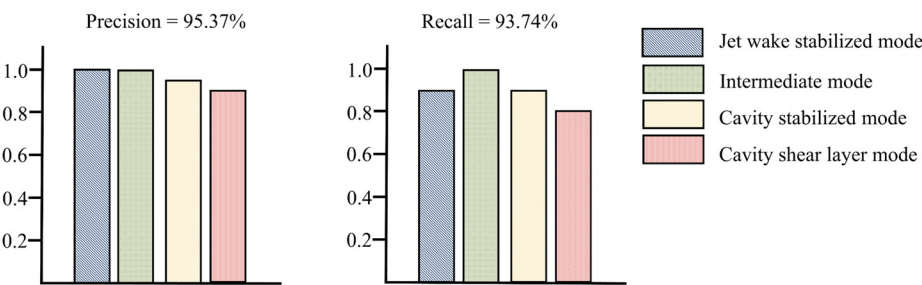


FIG. 7. Precision and recall of mode recognition.

Swin-T, VGG16 lags slightly in terms of convergence speed compared to the model adopted in this work. Specifically, VGG16’s loss stabilizes around 17 epochs. The remaining models perform inferior to Swin-T in both loss value and convergence speed. ResNet has the highest loss among them, while MobileNet exhibits the slowest convergence speed with significant oscillations in its loss curve. The performance of ViT lies between these two extremes.

The data in Table III show the performance of different models in terms of accuracy, precision, recall, and training time. It can be seen that Swin-T achieves the best performance in all evaluation metrics except recall, demonstrating its performance advantage among the various models. Although ResNet 50 has the highest recall rate, it requires twice the training time of Swin-T. Additionally, it lags behind Swin-T by 0.9% in accuracy and 2.65% in precision. In comparison, Swin-T exhibits superior overall performance. The performance of VGG16 is most comparable to Swin-T, with differences of less than 1% across all evaluation metrics except training time. However, VGG16 demands the longest training duration. Although Swin-T is inspired by the design of ViT, ViT’s overall performance is the poorest. Except for a slightly shorter training time compared to VGG16, ViT scores the lowest in all other metrics. This indicates that while the attention mechanism helps the model focus on important features, the cross-regional associative information is crucial for the data used in this study.

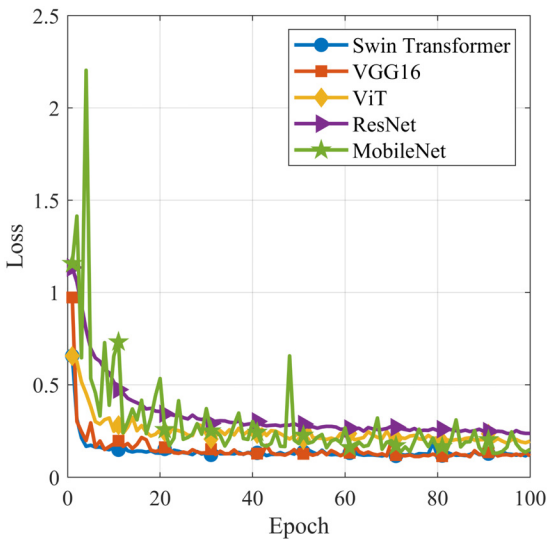


FIG. 8. Validation loss of different models with different epochs.

To assess the recognition performance of different models across various modes, it is crucial to present the precision and recall metrics for each model. Figure 9 illustrates the detailed performance of five distinct models across various modalities. In terms of precision, Swin-T demonstrates superior performance, particularly in identifying the cavity shear layer mode. Although Swin-T slightly underperforms compared to ResNet in the cavity stabilized mode, it surpasses ResNet in recognition across other modes. The performance of other CNN-based networks is consistently inferior to that of Swin-T, underscoring the superiority of the model employed in this study. Regarding recall, Swin-T exhibits a significant advantage in both the intermediate and cavity stabilized modes, indicating its superior ability to identify features associated with these modes compared to traditional models. Despite weaker performance in recognizing the cavity shear layer mode, the overall performance of Swin-T underscores its efficacy as a flame mode recognition model.

D. Robustness test

In the combustion process, the intermediate products of the reaction and the oil mist or smoke produced by incomplete combustion affect the image quality severely.⁴⁷ By adding Gaussian noise to the original flame image data, the simulation of oil mist and smoke produced by heavy oil combustion can be achieved, thereby verifying the robustness of the model and testing its recognition capability under poor image quality conditions. The process of noise adding can be described as follows:

α′ = α + στ, (13)

where α′ represents the noisy image and α represents the original image. σ is the noise level and τ is a normal random distribution within a range of -2.576–2.576 with 99% probability. In the data preparing process, different noise levels are added to the original images separately. The noise level σ in the training dataset is changed from 0

TABLE III. Overall performance of the tested models. Boldface denotes results of the current Swin-T algorithm.

Model	Accuracy	Precision	Recall	Time of training
Swin-T	95.28%	95.37%	93.74%	246 min
ViT	92.57%	90.87%	92.40%	1365 min
VGG16	94.44%	94.83%	93.36%	3321 min
ResNet 50	94.38%	92.72%	95.00%	505 min
MobileNet	94.90%	94.43%	93.98%	290 min

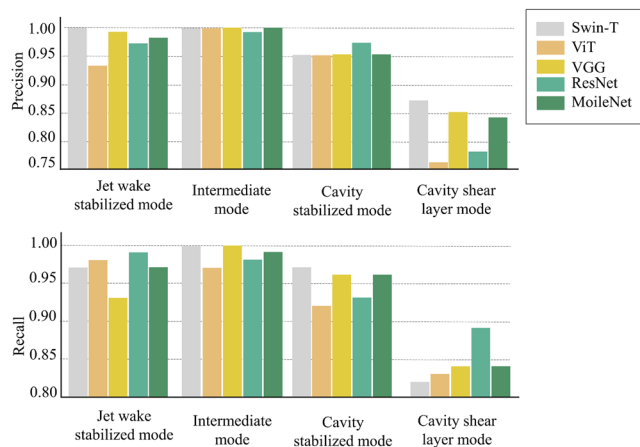


FIG. 9. Precision and recall of the tested models.

to 0.5 with a step size of 0.1. The model is then trained using data with varying degrees of noise. Similarly, Gaussian noise with different noise levels is also added to the validation datasets to investigate the robust performance of the Swin-T model. The corruption ratio in the validation dataset is also changed from 0 to 0.3 with a step of 0.1. In each testing phase, three trials were carried out and their average accuracy is shown in Fig. 10.

It can be observed from Fig. 10 that when the proportion of noise mixed into the training dataset is low (0, 0.1, 0.2), the accuracy of model decreases as the noise level in the validation dataset increases. The less noise mixed into the training data, the more pronounced the impact of noise in the test dataset. The highest prediction accuracy of the model is achieved by 95.28% (purple line in the figure) when $\delta = 0$. When the proportion of noise mixed into the training data exceeds 0.2, the accuracy stabilizes at around 92%. Even when a high proportion of noise is added to the validation dataset, the accuracy

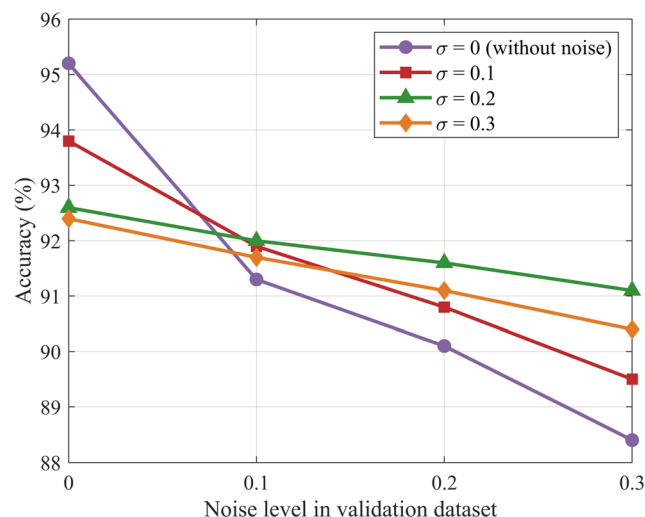


FIG. 10. Accuracy of the Swin-T model under different noise levels in the training and validation datasets.

remains stable. Therefore, robustness capability and high-accuracy performance are demonstrated through Swin-T. The results further demonstrate that, despite the rapid decline in accuracy as noise levels increase in the validation dataset, $\delta = 0.2$ consistently achieves the highest accuracy.

As Fig. 11 shows, in the tested models, Swin-T demonstrated a significant advantage in robustness. Compared to VGG16 and ResNet, the gap in robustness is considerable, with accuracy differences of 2.5% and 6.6% at noise level $\delta = 0.2$, respectively. Although MobileNet outperformed the model reported in this paper in terms of robustness in certain cases, the difference between them remained less than 1% across all tested scenarios. This difference stems from MobileNet's specialized design for improving robustness, as it utilizes scaling parameters for multi-scale feature extraction. For specific details, please refer to Howard *et al.*⁴⁸ These results indicate that the model reported in this paper has a certain advantage in robustness compared to other CNN models.

V. CONCLUSION

This paper presents a deep learning model, Swin Transformer, for the recognition of supersonic flame modes. The data are obtained from the scramjet cavity with the MCGA plasma experimental platform, where kerosene is used as fuel. Supersonic flame images are captured using a high-speed camera to acquire information about the flame structure. We investigated the appropriate structure of Swin-T, specifically focusing on Layer number and Projected dimension, to assess their impact on model size, computational resource consumption, and performance. The proposed method demonstrates high accuracy (95.28%), high precision (95.37%), high recall (93.74%), and low computational resource consumption and short training time. Furthermore, the model was compared with traditional recognition models, and its overall performance was superior to that of CNN-based recognition models. Additionally, by training and testing the model with data containing noise, we discussed the model's robustness and generalization capability. The results indicate that the model demonstrates good adaptability to noise and exhibits strong robustness,

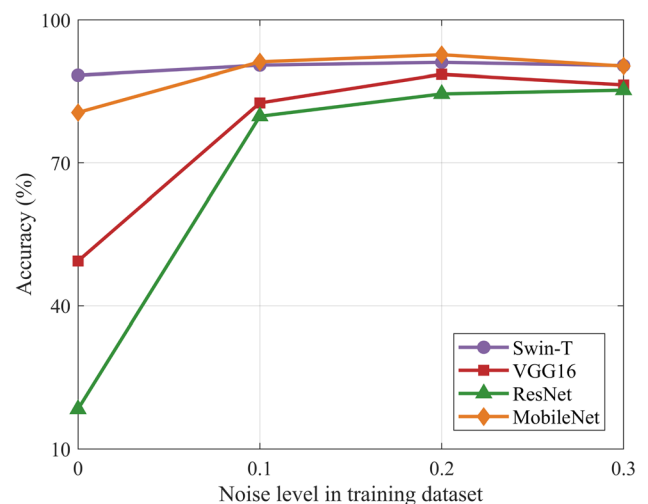


FIG. 11. Accuracy of the tested model under different noise levels in the training dataset.

suggesting that it can be applied to a wide range of combustion scenarios. The model's recognition results were also discussed, showing that it exhibits high confidence across all four modes. This indicates that the model has strong performance in feature extraction and analysis of the supersonic combustion modes.

Follow-up research will focus on developing more advanced models to further enhance both accuracy and real-time processing performance. These improvements aim to address the challenges of complex data dynamics and computational efficiency in practical applications. Additionally, we plan to extend the application of the proposed model to various combustion scenarios to rigorously evaluate its generalization capability. More explorations will provide valuable insights into the model's robustness and adaptability, contributing to its potential deployment in diverse real-world environments.

ACKNOWLEDGMENTS

This study was supported by the National Key Research and Development Program of China under Grant No. 2023YFC3107200, the National Natural Science Foundation of China under Grant No. 52376160, the National Natural Science Foundation of China under Grant No. 52006137, and the National Postdoctoral Researchers Program of China under Grant No. GZC20231559.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Qinlin Guan: Conceptualization (lead); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Resources (equal); Software (lead); Validation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead). **Xue Dong:** Funding acquisition (equal); Project administration (equal); Resources (equal); Supervision (equal); Writing – review & editing (equal). **Wei Zhang:** Formal analysis (equal); Methodology (equal); Software (equal); Supervision (lead); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (lead). **Liyang Sun:** Formal analysis (equal); Software (lead); Visualization (equal); Writing – original draft (equal). **Jiajian Zhu:** Conceptualization (equal); Data curation (equal); Funding acquisition (equal); Project administration (equal); Supervision (equal); Writing – review & editing (equal). **Tiangang Luo:** Conceptualization (equal); Data curation (equal); Methodology (equal); Supervision (equal); Visualization (equal). **Shuogui Xu:** Funding acquisition (lead); Project administration (equal); Supervision (supporting).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

NOMENCLATURE

ANN	Artificial neural networks
CNN	Convolutional neural network

DMD	Dynamic mode decomposition
ER	Equivalence ratio
Ma	Mach number
MCGA	Multi-channel gliding arc
ML	Machine learning
MLP	Multi-layer perceptron
MSA	Multi-head self-attention
m_a	Total incoming air mass flow rate, g/s
m_k	Kerosene mass flow rate, g/s
POD	Proper orthogonal decomposition
P_t	Total incoming air pressure, MPa
Swin-T	Swin Transformer
T_t	Total incoming air temperature, K
ViT	Vision transformer

APPENDIX: THE MULTI-HEAD ATTENTION MECHANISM

The transformer model³³ consists of multi-head self-attention layers. The self-attention mechanism can highlight faulty features with important information and suppress invalid features.^{49,50} Self-attention mechanism pays more attention to the internal correlation of sample data.⁵¹ The mathematical expressions of self-attention mechanism are

$$\begin{cases} \mathbf{Q} = \mathbf{W}^q \mathbf{X} \\ \mathbf{K} = \mathbf{W}^k \mathbf{X} \\ \mathbf{V} = \mathbf{W}^v \mathbf{X} \\ \text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right) \mathbf{V}, \end{cases} \quad (\text{A1})$$

where $\mathbf{X}$ denotes a spatial sequence vector representing input image data. Three different weight matrices $\mathbf{W}^q$, $\mathbf{W}^k$, and $\mathbf{W}^v$ are required in the linear mapping process of $\mathbf{X}$. Two information mappings, $\mathbf{Q}$ and $\mathbf{K}$, respectively, represent a certain part of the features extracted from the image and the similarity between the positions corresponding to those extracted from various parts of the image. $\mathbf{V}$ contains the weight information, and this matrix is used to determine the importance of features in the image. The d is specified when mapping the input vector to the $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ matrices through linear transformation. $\sqrt{d}$ is used as a normalization term in the self-attention mechanism. To ensure that more representative features are considered during the transformation, the multi-head self-attention (MSA) utilizes multiple linear projections to map $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ into corresponding dimensions, enabling the utilization of the most recently learned information in each execution process. Then, the output from the process is concatenated and projected to obtain the final result. The whole process can be expressed as follows:

$$\begin{cases} \text{head}_i = \text{Attention}(\mathbf{Q}\mathbf{W}_i^q, \mathbf{K}\mathbf{W}_i^k, \mathbf{V}\mathbf{W}_i^v) \\ \text{Multihead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) \mathbf{W}_o, \end{cases} \quad (\text{A2})$$

where head_i is the linear projection times of $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ and it will compute the respective self-attention score, and the information learned from different heads is combined by *Concat*.

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